
Education

2023.09 – Present **Ph.D. & M.S. in Computational Mathematics, Stanford University**

Thesis The exchange-correlation functional: exact definition, computational access, and structured approximation.

Advisor Prof. Lexing Ying

2019.10 – 2023.08 **Ph.D. & M.S. in Chemical Physics, University of Chicago**

Thesis Multiscale methods for quantum many-body systems.

Advisor Prof. Yuehaw Khoo

2015.09 – 2019.07 **B.S. in Physics, University of Chinese Academy of Sciences**

Thesis Density matrix renormalization group applied to quantum-chemical calculations.

Advisor Prof. Tao Xiang

2015.09 – 2019.07 **B.S. in Chemistry, University of Chinese Academy of Sciences**

Thesis GPU acceleration of matrix-product-state-based hierarchical equations of motion.

Advisor Prof. Qiang Shi

Research Experience

2025.06 – Present **Research Intern, ByteDance Ltd.**

Project Robust machine-learning quantum-chemistry methods for molecular science.

Advisors Dr. Yixiao Chen and Dr. Weiluo Ren

2021.06 – 2021.08 **Research Intern, Flatiron Institute, Center for Computational Quantum Physics**

Project Low-rank Green's function representations applied to dynamical mean-field theory.

Advisors Dr. Jason Kaye, Dr. Kun Chen, and Dr. Olivier Parcollet

Preprints and Manuscripts

2026 Titouan Duston, Jiashu Liang, Yuanheng Wang, Weihao Gao, Xuelan Wen, **Nan Sheng**, Weiluo Ren, Yang Sun, and Yixiao Chen. *Agentic discovery of exchange-correlation density functionals*. arXiv:2605.05460.

2026 **Nan Sheng**. *A density-functional perspective on force fields*. arXiv:2604.25215.

2026 **Nan Sheng**. *A unified variational framework for the inverse Kohn–Sham problem*. arXiv:2603.23452.

2026 **Nan Sheng**. *Exact density-functional theory as parallel ensemble variational hierarchies: from Lieb's formulation to Kohn–Sham theory*. arXiv:2603.23399.

2025 Titouan Duston, Shuo Xin, Yang Sun, Daoguang Zan, Aoyan Li, Shulin Xin, Kai Shen, Yixiao Chen, Qiming Sun, **Nan Sheng**, Ge Zhang, et al. *AlnsteinBench: benchmarking coding agents on scientific repositories*. arXiv:2512.21373.

2025 **Nan Sheng**, Xun Tang, Haoxuan Chen, and Lexing Ying. *Approximation of high-dimensional Gibbs distributions with functional hierarchical tensors*. arXiv:2501.17143.

Journal Articles

2025 Xun Tang, **Nan Sheng**, and Lexing Ying. *Solving high-dimensional Hamilton–Jacobi–Bellman equations with functional hierarchical tensor*. *SIAM Journal on Scientific Computing*, to appear. arXiv:2408.04209.

2023 **Nan Sheng**, Alexander Hampel, Sophie Beck, Olivier Parcollet, Nils Wentzell, Jason Kaye,

- 2020 He Ma, **Nan Sheng**, Marco Govoni, and Giulia Galli. *First-principles studies of strongly correlated states in defect spin qubits in diamond*. *Physical Chemistry Chemical Physics* 22, 25522–25527. doi:10.1039/D0CP04585C.

Talks

- 2025 Field-theoretic and tensor-network methods for scientific computing, Seed Research Intern Seminar, *ByteDance Ltd.*
- 2024 Introduction to computational quantum physics, Workshop on Applied Math for Quantum Physics, *University of Chicago*.
- 2023 Introduction to computational quantum physics, Student Applied Math Seminar, *Ohio State University*.
- 2023 Green's function formulation of quantum defect embedding theory, *APS March Meeting*.
- 2023 Quantum simulations of fermionic Hamiltonians with efficient encoding and ansatz schemes, *APS March Meeting*.
- 2022 Extrinsic and intrinsic defects in MgO and CaO as potential spin-qubit candidates, *APS March Meeting*.
- 2022 An exact double-counting scheme for quantum defect embedding theory, *APS March Meeting*.
- 2021 Accelerating dynamical mean-field calculations using the discrete Lehmann representation, CCQ Summer Intern Seminar, *Flatiron Institute*.
- 2021 First-principles studies of strongly correlated states in defect spin qubits in diamond, *APS March Meeting*.
- 2021 Coupling interoperable software for quantum simulations of materials, *APS March Meeting*.

Academic Service

- Reviewer PRL, PRX, PRX Quantum, PRB, PRA, PR Research, Chinese Physics B, Computer Physics Communications, Physics of Fluids, MLST, Quantum Reports, and PCCP.

Teaching

- Spring 2026 **CME 100: Vector Calculus for Engineers**, *Stanford University*, Course Assistant
- Fall 2025; Winter 2026 **MATH 19: Calculus**, *Stanford University*, Course Assistant
- Winter 2025 **CME 306: Computational Methods of Applied Mathematics**, *Stanford University*, Course Assistant
- Spring 2020 **CHEM 11300: Comprehensive General Chemistry III**, *University of Chicago*, Teaching Assistant
- Fall 2019; Winter 2020 **CHEM 22000/22100: Organic Chemistry I/II**, *University of Chicago*, Teaching Assistant

Fellowships and Awards

- 2023 **ICME Fellowship**, *Stanford University*
- 2019 **McCormick Fellowship**, *University of Chicago*
- 2019 **Excellent Graduate of Beijing**, *Chinese Ministry of Education*, 2 out of 35
- 2019 **Excellent Graduate**, *University of Chinese Academy of Sciences*, 3 out of 35
- 2018 **Study Abroad Scholarship**, *University of Chinese Academy of Sciences*, 2 out of 35
- 2018 **Tang Lixin Scholarship**, *University of Chinese Academy of Sciences*, 1 out of 35
- 2016, 2017, 2018 **National Encouragement Scholarship**, *Chinese Ministry of Education*, 2 out of 35

2016, 2017, 2018 **Academic Excellence Scholarship**, *University of Chinese Academy of Sciences*

Technical Skills

Research	Computational science; numerical methods; mathematical modeling; optimization; machine learning; electronic structure.
Programming	Python, MATLAB, Fortran, C/C++, Bash, $\LaTeX$, MPI, CUDA.
Software	CVXPY, JAX, PySCF, Qiskit, Quantum ESPRESSO, Gaussian, ORCA, TRIQS, Wannier90.